



Using BOE 4.3 with RDM DevOps and Azure

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USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Configuring Crystal Reports in RDM DevOps and Azure

Prerequisite: You require the BusinessObjects Client Tools 4.3 including Crystal Reports 2020 version to work with BOE 4.3 environments.

Before we can deploy a report to an Environment we must configure the RDM DevOps Environment and the Azure DevOps environment.

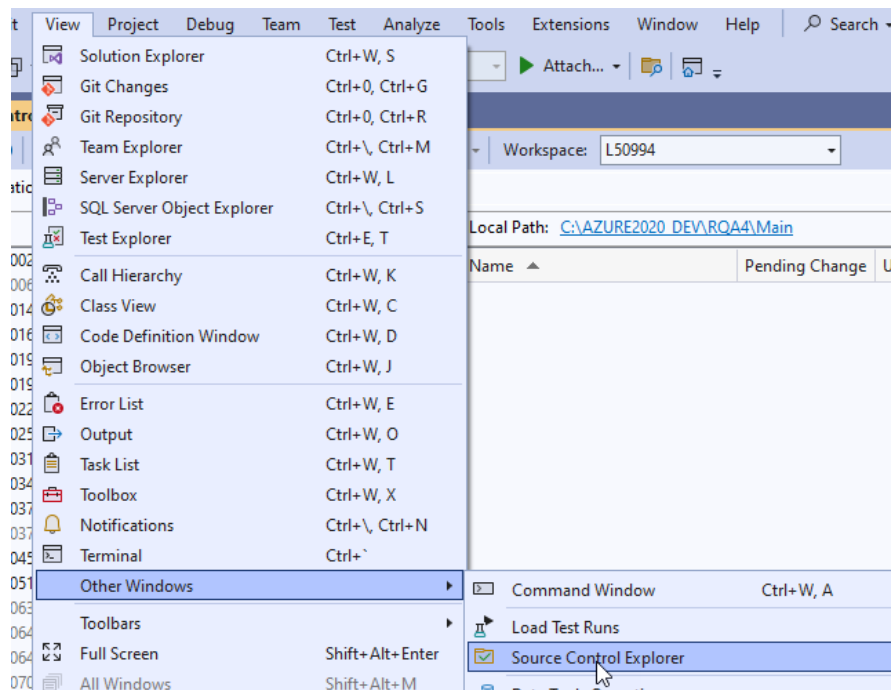
Task Summary

- 1- Add Crystal Reports folder to Azure DevOps
- 2- Add Crystal Reports component in RDM DevOps under your project
- 3- Add a new ODBC component in RDM DevOps
- 4- Configure ODBC component in RDM DevOps

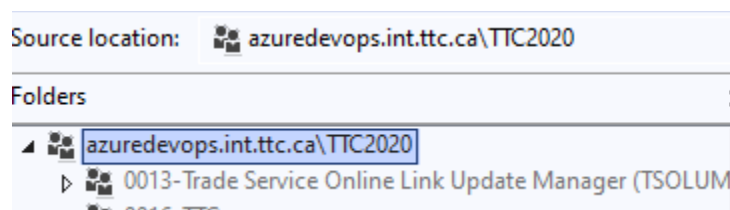
Detailed Steps

1- Add Crystal Reports folder to Azure DevOps

- a. Open Visual Studio and connect to Azure <https://azureddevops.int.ttc.ca/TTC2020> and select from the menu bar **VIEW->OtherWindows-> Source Control Explorer**

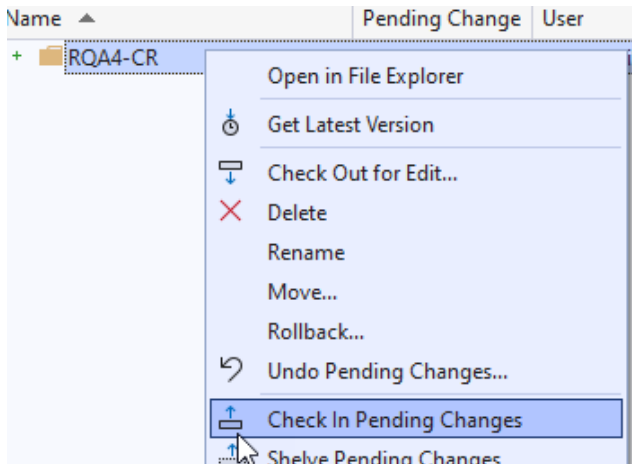
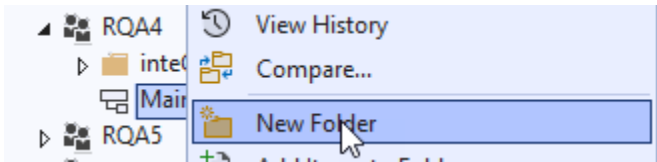


- b. Select your project folder that you will be working in. If this has not been created, please contact your Azure DevOps admin and/or project owner. You will have to open a request to have your project created.



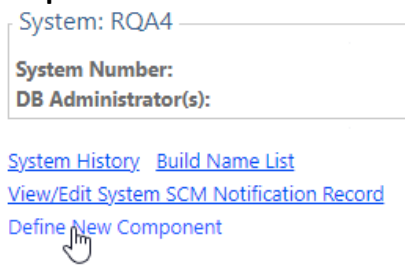
USING BOE 4.3 WITH RDM DEVOPS AND AZURE

- c. Create a folder in your project that will hold all your crystal reports. It will take the form as follows **<name>-CR**. Right click on the **Main** branch and select **New Folder** for this example we will use **RQA4-CR**. Check in newly created folder under **Main** branch.

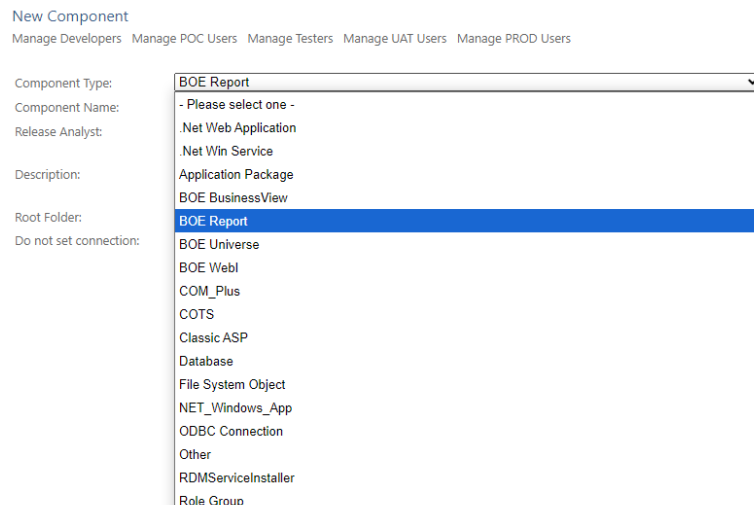


2- Add Crystal Reports component in RDM DevOps under your project

- a. In your project, select **Define New Component**



- b. Select the dropdown menu under **Component Type** and select **BOE Report**.



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- c. Use the folder you named in step 1c, in this example it would be **RQA4-CR**. This folder appears in the dropdown for **Component Name**.

New Component

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Component Type:	BOE Report
Component Name:	RQA4-CR
Release Analyst:	- Please select one - RQA4-CR
Description:	
Root Folder:	
Do not set connection:	☐

[Save](#) [Return](#)

- d. Enter the **description** of the component and the **root folder**. Leave **Do not set connection** checkbox unchecked if you want RDM DevOps to automatically set connection when deploying. Click **Save**.

New Component

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Component Type:	BOE Report
Component Name:	RQA4-CR
Release Analyst:	BOEAdmin
Description:	BOE Report Component for testing BOE Reports deployments
Root Folder:	RQA4-CR
Do not set connection:	☐

[Save](#) [Return](#)

Note: The majority of the root folders will be the same name as your component since the component name comes from the folder name you created in Azure DevOps source control under the **Main** branch.

- e. Click **Create Dev Environment**.

System: RQA4		Project Manager: caguerra			
System Number: 1014	DB Administrator(s): xizheng	Lead Developer(s): m/aciachshnrv			
System History Build Name List View/Edit System SCM Notification Record Define New Component					
Component Name	DEV	POC	TST	UAT	PROD
RQA4-CR					
Dev Lead's Task		QA Lead's Task		Admin's Task	
Create Dev Environment Create POC Environment SCM Structure Define Build Name Deploy to Dev Deploy to POC Ready for QA Testing Ready for Production		QA Lead(s) Create Test Environment Deploy to Test Ready for UAT Create UAT Environment Deploy to UAT Ready for Production Reject Build		Create Production Environment Deploy to Production Release Notification Promote to Production System Management Manage Build	

- f. Select component and click **Create Environment**.

Create DEV Environment for System: RQA4

Component	Existing	Component Type
RQA4-CR	☒	BOE Report
Create Environment Cancel		

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g. Environment is created.

Component Name	DEV
<u>RQA4-CR</u>	(BOE)

Dev Lead's Task

Create Dev Environment

QA Lead's Task

[QA Lead\(s\)](#)

h. If you click on **BOE** link , it opens BOE dev at <https://boe.dev.int.ttc.ca/BOE/BI>



Note: The component will now appear in the component list in RDM DevOps. Also, email notification is sent.

RDMDevOps Notification - DEV Environment Created - RQA4 / RQA4-CR

 RdmDevOps.Web.Dev@ttc.ca
To  Gaddour, Hammam;  Erturk, Onder;  Udall, Mike;  Jourakovsky, Igor;  Vlacic, Nikolina;  Guerra, Carlos;  Vlacic, Nikolina;  Srinivasan, Shriram

Enterprise Vault

Create Environment Log: RQA4 / RQA4-CR

Actor: nvlacic

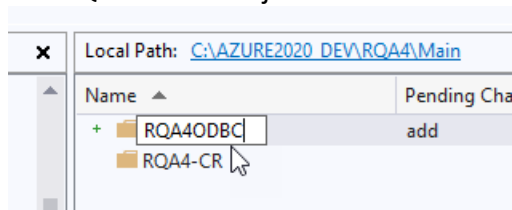
Date/Time: 2024-10-28 14:22

Details:

- Create Application Group RQA4-CRDEV
- Map Component Group intttcca\RQA4-CRDEV to BOE Environment
- Create BOE Application ID RQA4-CR_DEV_ID
- Create BOE Folder RQA4-CR.DEV
- Assign ViewOnDemand & Download Files permissions to BOE Group on the BOE Folder
- Assign ViewOnDemand & Download Files permissions to BOE Application ID on the BOE Folder

3- Add a new ODBC component in RDM DevOps

a. Create a folder in your project that will hold all your crystal reports. It will take the form as follows **<name>-ODBC**. Right click on the **Main** branch and select **New Folder** for this example we will use **RQA4**. Check in just added folder.



b. In your project, in RDM DevOps, select **Define New Component**

System: RQA4	
System Number:	1014
DB Administrator(s):	xizheng

[System History](#) [Build Name List](#)

[View/Edit System SCM Notification Record](#)

[Define New Component](#)

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- c. Select the dropdown menu under **Development Platform** and select **ODBC Connection**

New Component

Manage Developers Manage POC Users Manage Testers Manage UAT Users Manage PROD Users

Component Type: ODBC Connection

Component Name: - Please select one -

Release Analyst: - Please select one -

RQA4ODBC

- d. Enter TNS Name for Oracle connection type **OR** Database Name and Database Server or SQLServer connection type. Make sure you fill in tab for all environments: **DEV, POC, TST, UAT and PROD.**

New Component

Manage Developers Manage POC Users Manage Testers Manage UAT Users Manage PROD Users

Component Type: ODBC Connection

Component Name: RQA4ODBC

Release Analyst: ServerAdmin

Description: ODBC connection for testing project

ODBC Connection Type: Oracle

Data Source Name: BOE

ODBC Server Type: BOE

Environment: Dev POC Test UAT Production

TNS Name: rdmuoc_dev

Database Name:

Database Server:

Save Return

New Component

Manage Developers Manage POC Users Manage Testers Manage UAT Users Manage PROD Users

Component Type: ODBC Connection

Component Name: RQA4ODBC

Release Analyst: ServerAdmin

Description: ODBC connection for testing project

ODBC Connection Type: SQLServer

Data Source Name:

ODBC Server Type: BOE

Environment: Dev POC Test UAT Production

TNS Name:

Database Name: RDMUSC

Database Server: SQL2019DEV

Save Return

- e. Create ODBC connection on BOE Dev server by clicking on Create Dev Environment.

Create DEV Environment for System: RQA4

Component	Existing	Component Type
RQA4-CR	☒	BOE Report
RQA4ODBC	☒	ODBC Connection

Create Environment Cancel

- f. ODBC connection is created on DEV server. Create environments for added components on all other environments

System: RQA4

System Number: 1014

DB Administrator(s): xizheng

[System History](#) [Build Name List](#)

[View/Edit System SCM Notification Record](#)

[Define New Component](#)

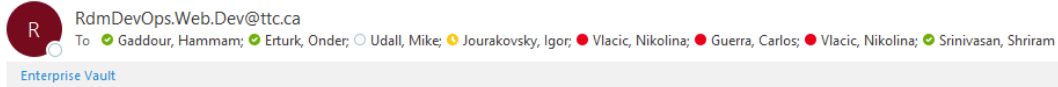
Component Name	DEV
RQA4-CR	(BOE)
RQA4ODBC	ODBC

Dev Lead's Task OA Lead's Task

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- g. The result will be an email from the RDM DevOps system indicating the component was created.

RDMDevOps Notification - DEV Environment Created - RQA4 / RQA4ODBC



Create Environment Log: RQA4 / RQA4ODBC

Actor: nvlacic

Date/Time: 2024-10-28 14:43

Details:

- SQL Server ODBC connection: created on D0073w001

- h. You need to contact DBA member to add your database connection to a system called ARIS. Connection needs to be entered in ARIS with **the same userID** used for reports for all environments.

Note: Please make sure that you have updated your local Oracle client to 64 bit if you are using Oracle database when upgrading to BOE 4.3 as well as BOE 4.3 version of SAP Business Intelligence Client and Crystal Reports 2020. Local ODBC, when creating reports, needs to use the very same userID as it gets embedded inside report properties, being deployed in other environments so ARIS entry for each environment needs to have connection with the same user ID for all environments, used for reporting.

Business Views, used by reports also need to contain same user ID when looking for password in ARIS encrypted connection string.

Deploying Crystal Reports and Web Intelligence Reports to a specific Environment

Once we have our RDM DevOps and Azure project setup correctly we are ready to deploy some reports to any and all environments. There are five environments within the RDM DevOps application; they are DEV, POC, TST, UAT, and PROD.

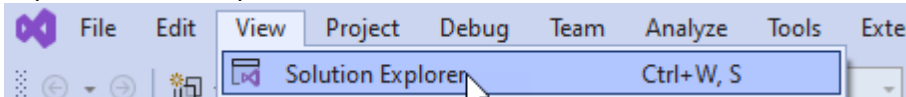
Task Summary

- 1- Apply a label to the Crystal Report folder in Azure DevOps report component folder
- 2- Create a version for RDM DevOps system using the newly created label
- 3- Deploy to an environment using the new build number you have created.

Detailed Steps

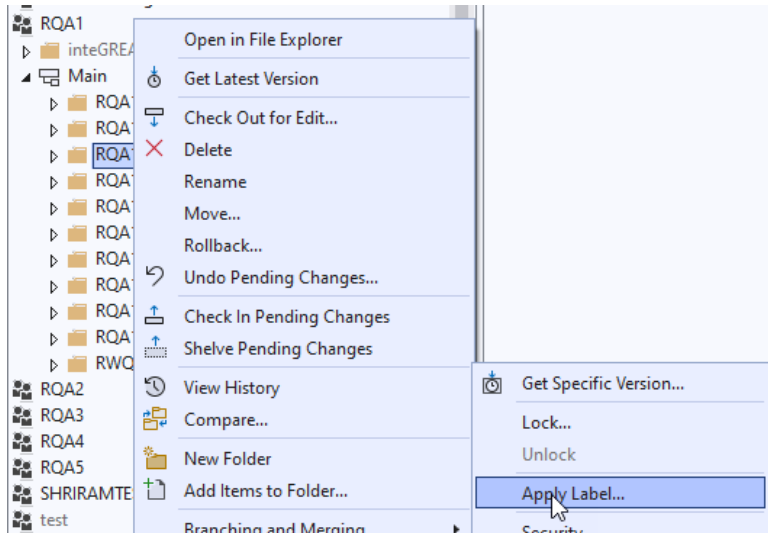
1- Apply a label to the Crystal Report folder in Azure

- a. Open AzureDevOps or Visual Studio and select from the menu bar **VIEW->Team Explorer**

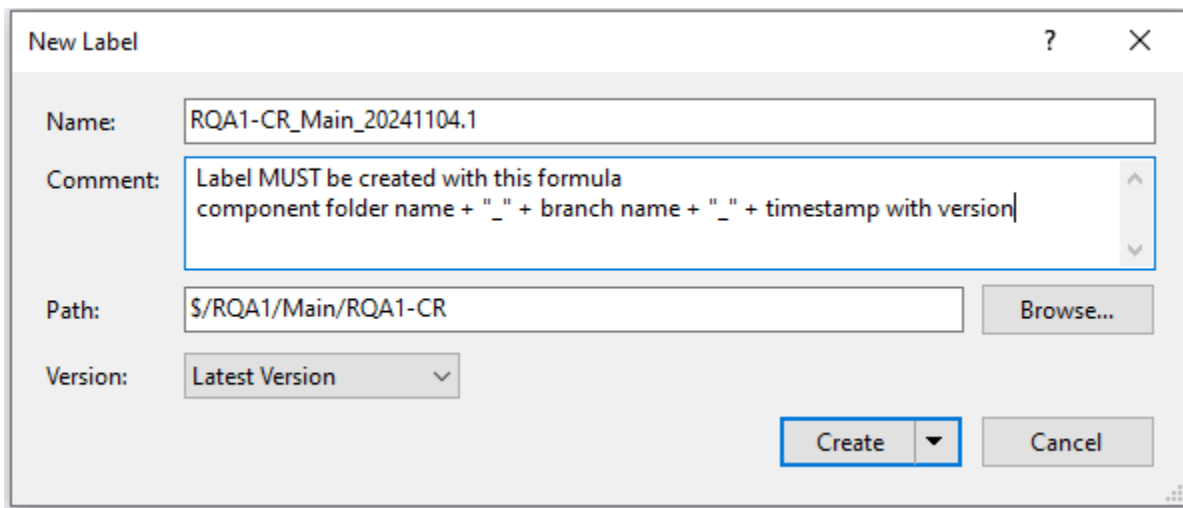


- b. Select your project folder that you will be working in. **Right click** on your Crystal Reports folder/component. This folder is usually suffixed with a **-CR** or **-CR31**. Select the option **Advanced->Apply Label**.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

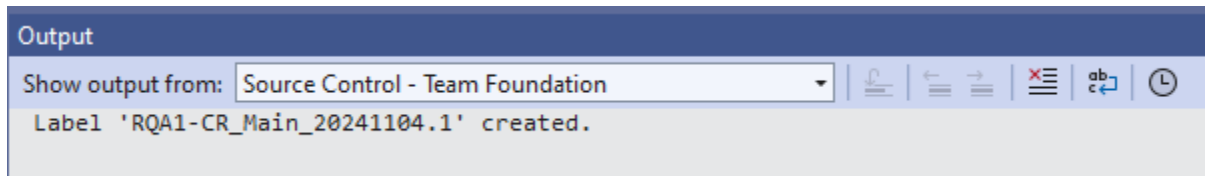


- c. On this screen you must enter the label you would like to name this component/folder. The format of label name is as follows: **<Component Name>_<Branch Name>_<YYYYMMDD>**
Example: RQA1-CR_Main_20241104.1



- d. Press the **Create** button.

In the output window of Visual Studio you will have confirmation that the label was created.



2- Create a Build In RDM DevOps using the newly created label

- a. Go to the RDM DevOps website and select your project from the list of projects on the main menu.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE



RDMDevOps

Release and Deployment Manager for Azure DevOps

Welcome **Vladic, N**

My System

Initiate System

Reports

System Search

Pending Tasks

User Guide..

Admin Tools

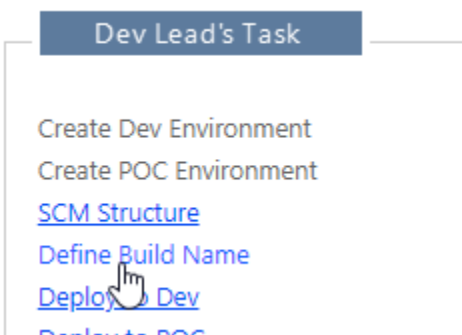
RDM DEVOPS - SYSTEM LIST

System Name:

Search

System Number	System Name	View History	View/Edit
0101	RQA1	System History	View/Edit

- b. Select **Define Build Name** from the menu under the **Dev Lead's Task**. It will bring you to the **Define Release Version for System** page shown below.



- c. The build name will increment by one automatically from the last digit of the last build name. If it's your first build with the project it will be 0.0.0.1. The details of what each number represents is **[Major].[Minor].[Revision].[Incremental]**. You can update the number manually if required to do a build that is not an Incremental build.

Select matching component with label you just created and click **Save**.

Define Release Version for System: RQA1

Build Name: RQA1 Version

Import from:

Build Name:

Reference Type:

Change Order Number: NOT NORMAL

Component Builds:

Component Name	TFS label
RQA1-CR	RQA1-CR_Main_20241104.1

No records to display.

3- Deploy to an environment using the new build number created.

- a. Go to the RDM DevOps website and select your system. As a developer, deployment can only be made in the DEV and POC environments. In this example we will deploy to the DEV environment. Select **Deploy to Dev**

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Dev Lead's Task

Create Dev Environment
Create POC Environment

[SCM Structure](#)

[Define Build Name](#)

[Deploy to Dev](#)

[Deploy to POC](#)

[Ready for QA Testing](#)

b. Select the release version you numbered earlier and click **Save**

Deploy to DEV for System: RQA1

By Release Version

By Build Label

By Saved Build Setting

Release Version:

RQA1 Version 1.0.0.3

☐ Show All

Component Name

TFS label

RQA1-CR

RQA1-CR_Main_20241104.1

Save

Cancel

For Web Intelligence (Webi), the process is equivalent.

Developers creates Webi reports using Web Intelligence Rich Client and saves created files to component folder to be deployed by RDM DevOps. Component folder is created under Main branch named **<name>-WEBI**. Select **BOE Webi** component type when creating new Webi component in RDM DevOps.

RDEV2 / RDEV2-WEBI

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Component Type:

BOE Webi

Component Name:

RDEV2-WEBI

Release Analyst:

BOEAdmin

Description:

test webi 4.3

Root Folder:

RDEV2-WEBI

Save

Return

Deploying a Webi component in RDM DevOps is the same as deploying Crystal Reports.

Dev Lead's Task

Create Dev Environment

Create POC Environment

[SCM Structure](#)

[Define Build Name](#)

[Deploy to Dev](#)

[Deploy to POC](#)

[Ready for QA Testing](#)

RDMDevOps Notification - Build and Deployment Log - RDEV2 (DEV)

 RdmDevOps.BuildEngine.Dev@ttc.ca
To: Vlacic, Nikolina

  Reply 

Build and Deployment Log: RDEV2 (DEV)

Actor: nvlacic

Build Server: D0195W003

Build Type: By Version

Build Name: RDEV2 Version 1.0.0.2

Change Order Number: 15287

Details:

Build Label(s):

RDEV2-WEBI_Main_20241030.1

- Deploy BOE Report(s) to D0073w001\RDEV2-WEBI.DEV
- Deployed RDEV2-WEBI to D0073w001

Setting Up Business Views in RDM DevOps and Azure

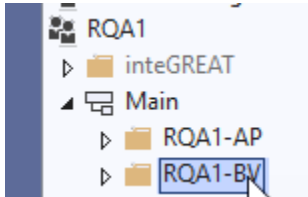
Prerequisite: You require the BusinessObjects Client Tools installed to work with business views.

Task Summary

- 1- Create a folder in your Azure project that will eventually store the business views. Usually the folder is names like <system acronym>-BV. **Example:** RQA1-BV, IFS-BV.
- 2- Create a Business View Component in RDM DevOps
- 3- Add developers the the DEV environment through RDM DevOps
- 4- Open the business view manager and create your business views

Detailed Steps

- 1- **Create a folder in your Azure project that will eventually store the business views. Usually the folder is names like <system acronym>-BV. Example: RQA1-BV, IFS-BV. Folder **MUST** be created under **Main** branch to be considered as component in RDM DevOps.**



- 2- **Create a Business View Component in RDM DevOps**
 - a. Click **Define New Component**

System: RQA1

System Number:	0101
DB Administrator(s):	xizheng

[System History](#)
[Build Name List](#)
[View/Edit System SCM Notification Record](#)
[Define New Component](#)

- b. Select **BOE v3.1 – Business Views** from the drop down list under Development Platform

New Component

[Manage Developers](#)
[Manage POC Users](#)
[Manage Testers](#)
[Manage UAT](#)

Component Type:	- Please select one -
Component Name:	- Please select one -
Release Analyst:	.Net Web Application
	.Net Win Service
	Application Package
	BOE BusinessView
	BOE Report

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Select the **Component Name** which should be the new folder you created in Azure DevOps, under **Main** branch. Type in the folder name from Azure DevOps in the **Root Folder** field. Click the **Save** button

RQA1 / RQA1-BV

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Component Type:	BOE BusinessView
Component Name:	RQA1-BV
Release Analyst:	BOEAdmin
Description:	Test BOE43
BOEBVFolder:	RQA1-BV

Save

Return

3- Add developers the the DEV environment through RDM DevOps

Click on your BV component link to open component settings.

System: RQA1
System Number:
DB Administrator(s):

[System History](#) [Build Name List](#)
[View/Edit System SCM Notification Record](#)
[Define New Component](#)

Component Name
RQA1-BV
RQA1-CR

If DEV environment is created, **Manage Developers** link is enabled, etc.

Note: DEV environment must be created in order for AD group to be created so it can be managed from the following screen.

Click **Manage Developers**

RQA1 / RQA1-BV

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Add users by selecting from list of ITS users or typing their windows user ID and clicking **Add**.

Note: The users must be in the ITS Users global group

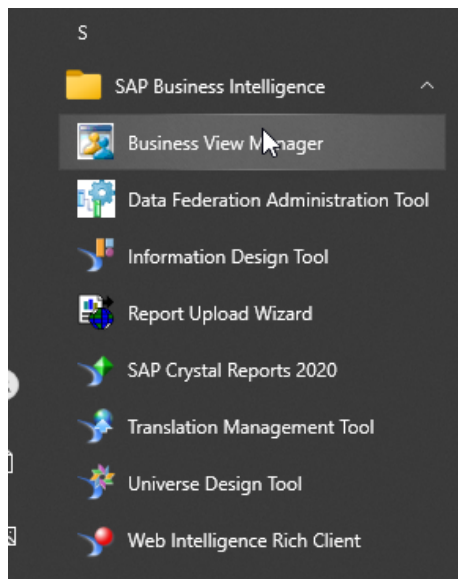
USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Manage DEV Group - System: RQA1 / Component: RQA1-BV

User's Account:

Remove	
Remove	Abiodun, Anne
Remove	Adams, Claire
Remove	Adhikari, Pranay
Remove	Aditya Jaiswal, Kumar
Remove	Ahmed, Zakaria
Remove	Akhtar, Syed

- 4- **Open the business view manager and create your business views**, which needs to be installed in your workstation as part of SAP Business Intelligence client.



Select DEV server name with Windows Authentication and press **OK**.

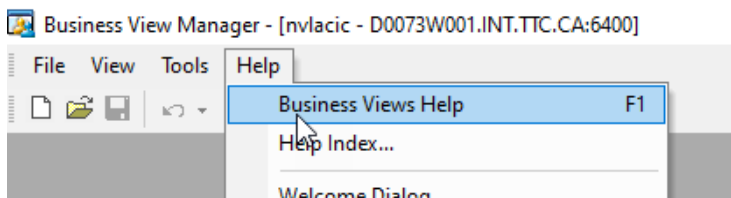
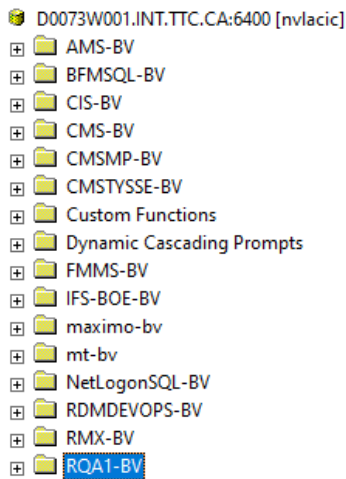
The dialog box is titled 'Log On to BusinessObjects Enterprise'. It contains the following fields and controls:

- System:** A dropdown menu with 'D0073W001' selected.
- User name:** An empty text input field.
- Password:** An empty password input field.
- Authentication:** A dropdown menu with 'Windows AD' selected.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom.

A mouse cursor is pointing at the 'OK' button.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Your folder for BV creation will be available and you can start creating Business Views for your system/reports. Business Views are created manually in DEV environment by developer.



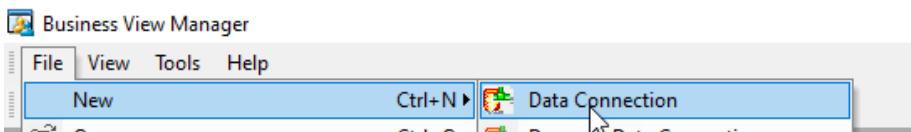
Creating Business Views in DEV environment using Business Views Manager

When RDM DevOps creates folder, it creates AD group which gets granted permission to BV folder on BOE server to create Business Views manually under that component folder.

Note: You need to have 64-bit ODBC connection set on your workstation for connection that you intend to use in business views. User ID for database connection used for Business View creation, needs to be set in ARIS with connection string for all environments: DEV, POC, TST, UAT, and PROD. Use the same connection and user in Business View and the Crystal Report component with the same User ID to make sure that both are connection to the same physical database and connection can be set automatically when deploying using RDM DevOps.

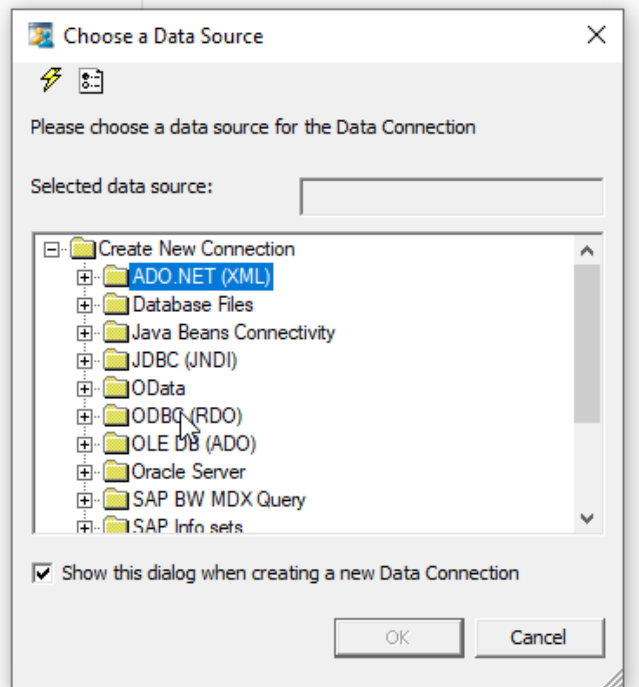
1- Create Data Connection

Click **New** and select **Data Connection**

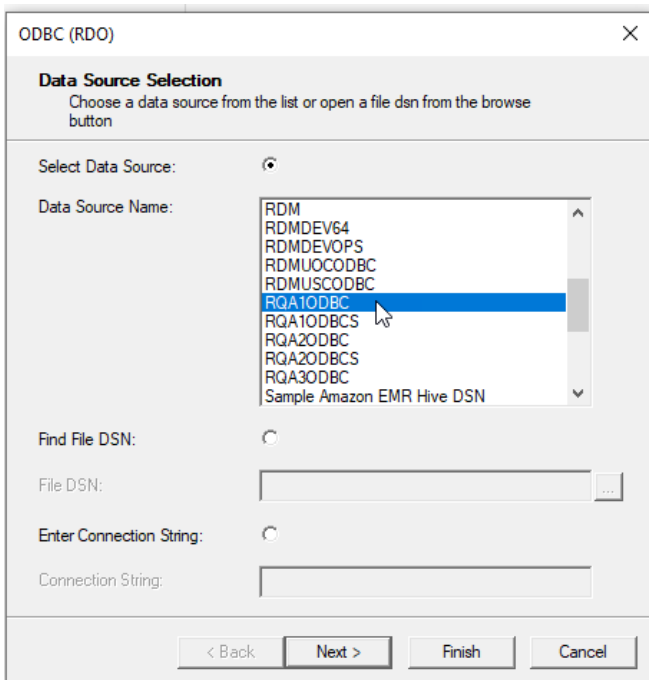


USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Choose Data Source ODBC (RDO). If using Oracle database, make sure that Oracle 64-bit client is installed in your workstation. SDO package is available to order in Service Catalogue. Crystal Reports 2020 is recommended for working with SAP BOE 4.3 Client Tools.

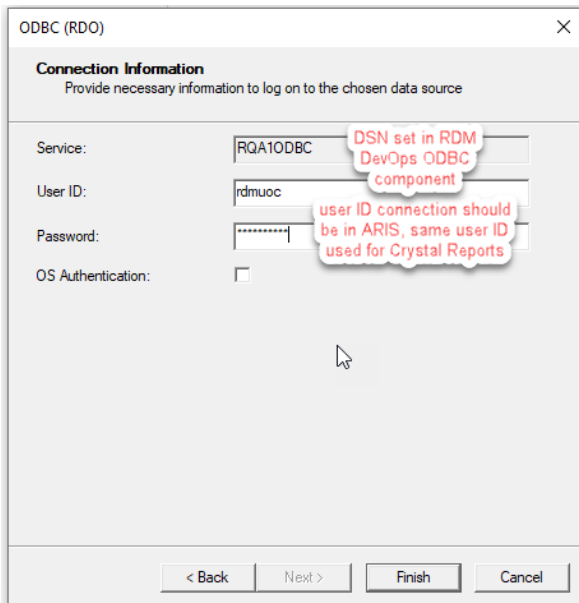


64-bit ODBC connection, saved in your workstation will show as options to choose from.



Click 'Next' and enter DB credentials.

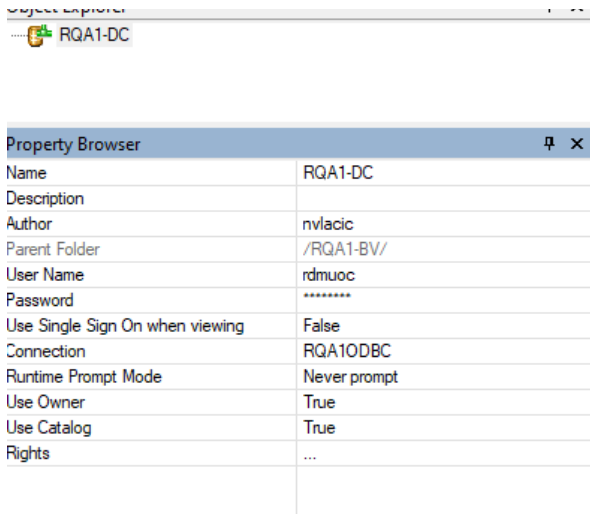
USING BOE 4.3 WITH RDM DEVOPS AND AZURE



ODBC (RDO) dialog box showing Connection Information. The Service is RQA10DBC, User ID is rdmuoc, and Password is masked. OS Authentication is unchecked. Red callouts indicate: 'DSN set in RDM DevOps ODBC component' pointing to the Service field, and 'user ID connection should be in ARIS, same user ID used for Crystal Reports' pointing to the User ID field.

Field	Value
Service	RQA10DBC
User ID	rdmuoc
Password	*****
OS Authentication	☐

Rename connection as **<system>-DC (RQA1-DC)** and save under your Business View folder (**RQA1-BV**). Make sure you select **Never Prompt** for **Runtime Prompt Mode** and enter same User Name as your ARIS connection entry, used for reports as well for all environments.

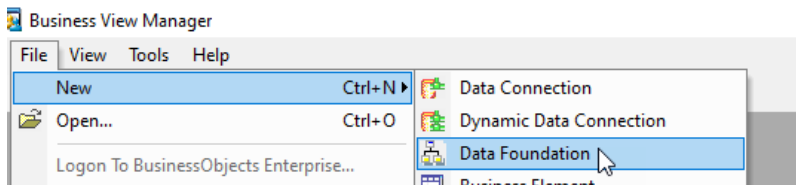


Property Browser for RQA1-DC. The table shows various properties and their values.

Property	Value
Name	RQA1-DC
Description	
Author	nvlacic
Parent Folder	/RQA1-BV/
User Name	rdmuoc
Password	*****
Use Single Sign On when viewing	False
Connection	RQA10DBC
Runtime Prompt Mode	Never prompt
Use Owner	True
Use Catalog	True
Rights	...

2- Create Data Foundation

Once you have entered Data Connection, you can create Data Foundation



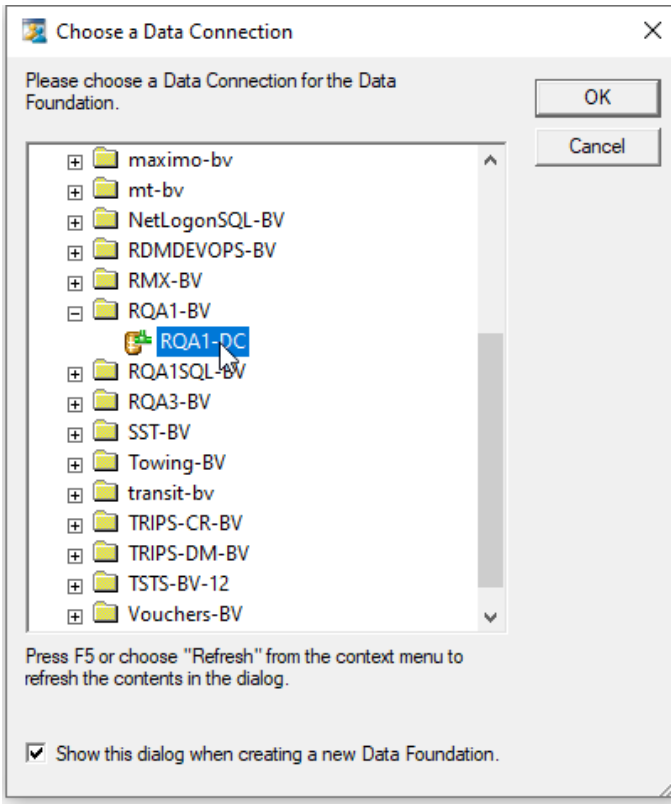
Business View Manager menu showing the 'New' option selected, with a submenu open showing 'Data Connection', 'Dynamic Data Connection', 'Data Foundation', and 'Business Element'.

Menu Item	Shortcut
New	Ctrl+N
Open...	Ctrl+O
Logon To BusinessObjects Enterprise...	

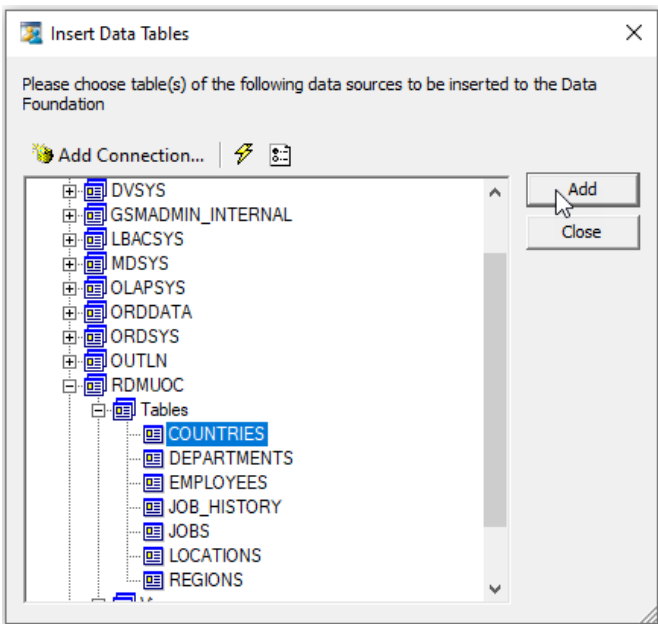
Submenu Item
Data Connection
Dynamic Data Connection
Data Foundation
Business Element

Select Data Connection you just created.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

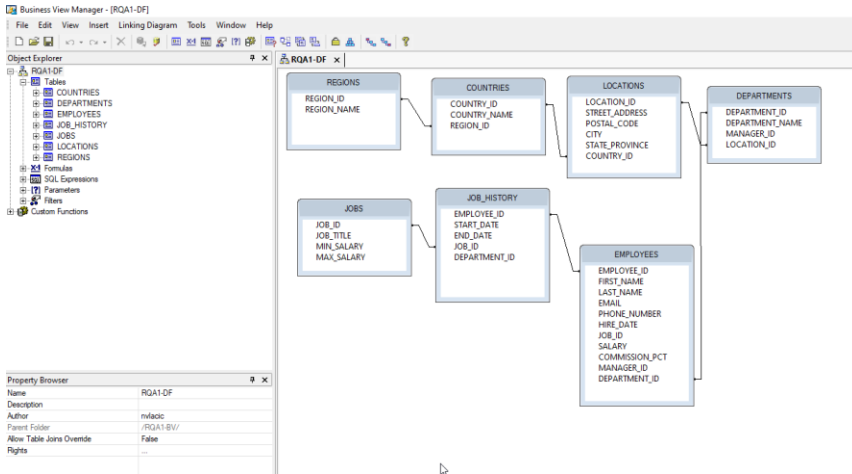


Select database tables from which you want to prompt data.



Name your Data Foundation as **<system>-DF (RQA1-DF)** and save under **RQA1-BV** folder you created using RDM DevOps. Tables included show all the fields you have available for creating list of values to choose from for dynamic prompt that you are creating.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

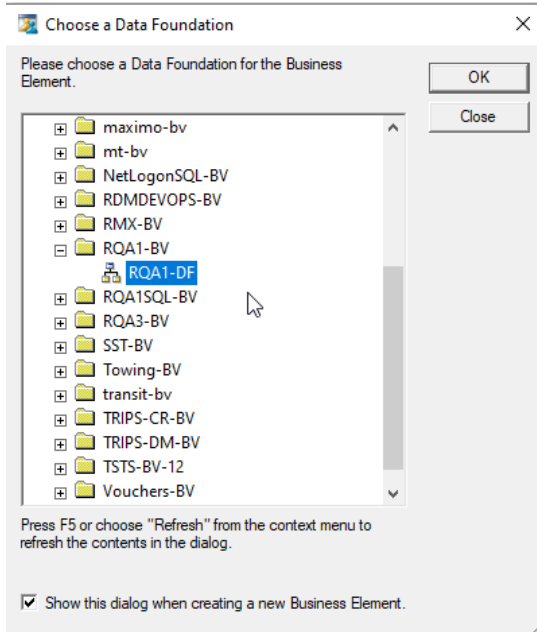


3- Create Business Element

Create New Business Element

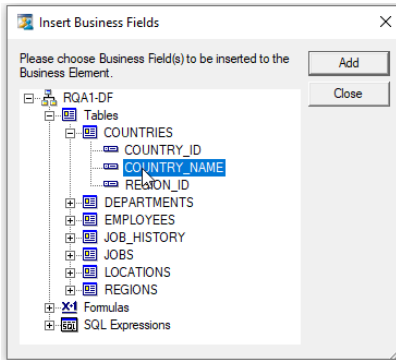


Choose Data Foundation you just created

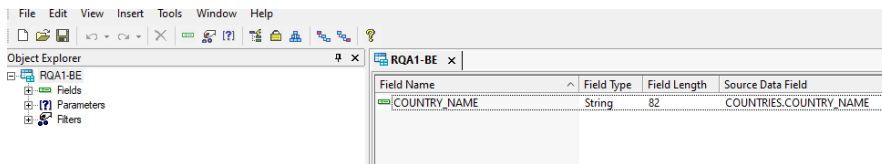


Select Table and column from which you are going to create list of values to load your dynamic prompt from.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

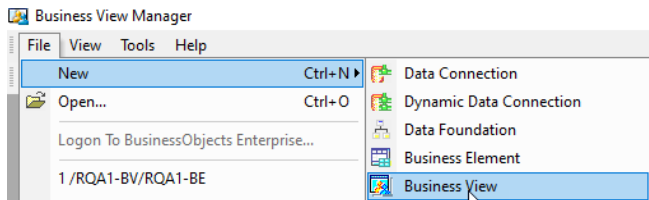


Click Add and rename your Business Element as **<system>-BE (RQA1-BE)** and same under your BV folder.

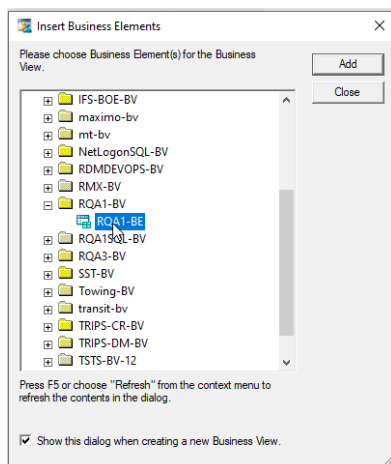


4- Create Business Views

Select create Business Views



Choose Business Element you just created

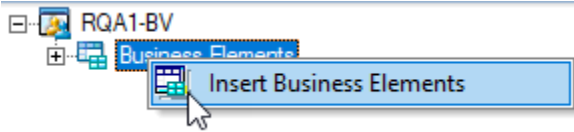


Save as **<system>-BV (RQA1-BV)** under the same RQA1-BV Business View folder.

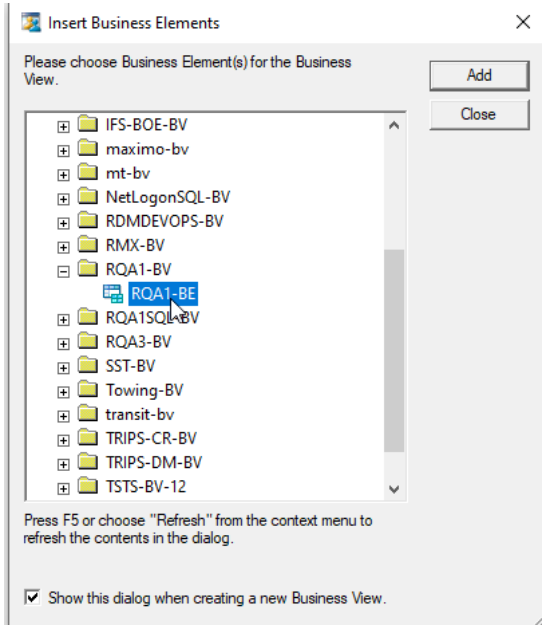
5- Add Business Element to a Business View

Open newly saved Business views and right click on Business Elements and select **Insert Business Elements**.

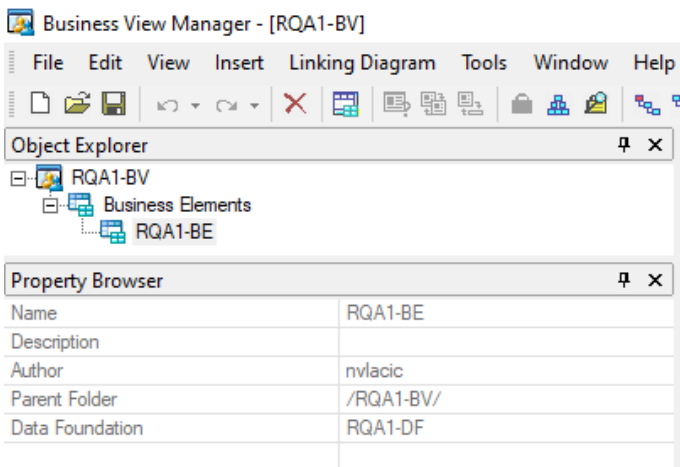
USING BOE 4.3 WITH RDM DEVOPS AND AZURE



Select Business Element that you just created. **Add** and **Save**.



Your Business Element shows in Object Explorer under your Business Views as on the screenshot below

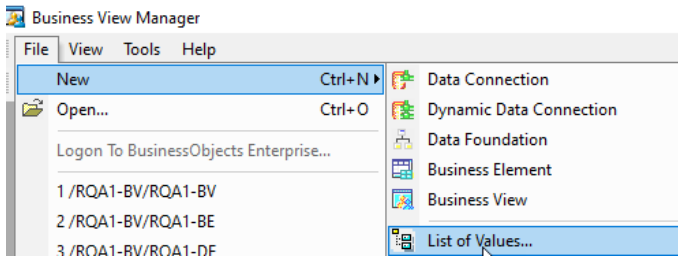


Now, you can create list of values, which can be chosen from Crystal Report to create Dynamic Prompt to be used by report.

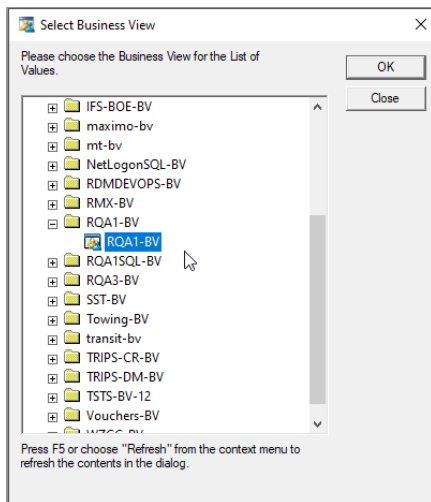
6- Create List of Values

Select New, then List of Values...

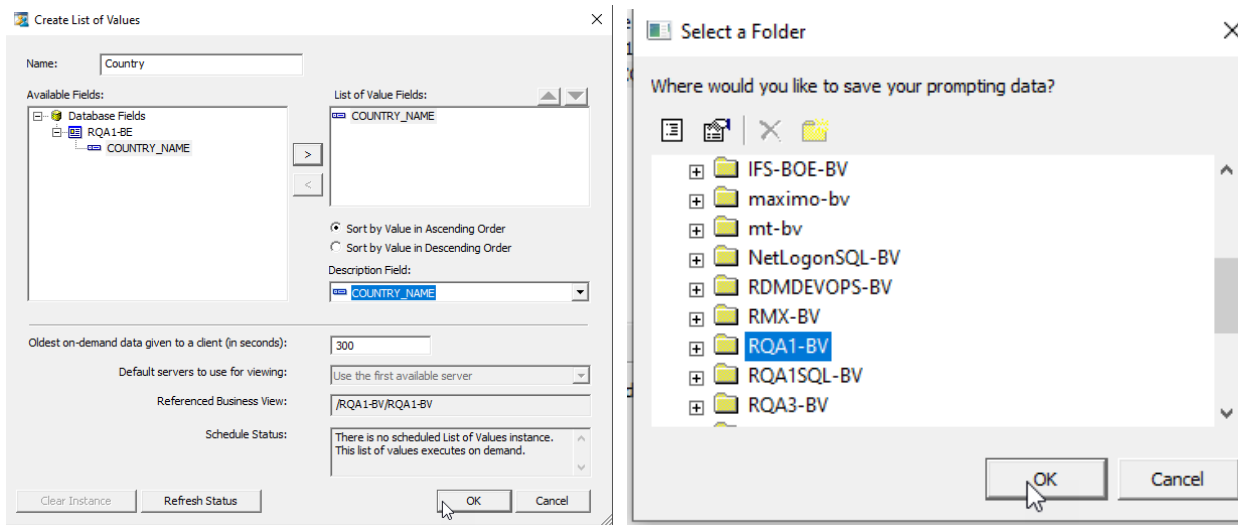
USING BOE 4.3 WITH RDM DEVOPS AND AZURE



Select Business View, from which to create it



Choose fields and click OK and select your BV component folder to save the list of values you just created.



7- Create Dynamic Prompt in your Business View

Now you can add dynamic prompt to your report parameter fields. **If this dynamic prompt is used for the first time, select list of values (see highlighted).** If not, use Prompt group to avoid of duplication of the same dynamic prompt on the BOE server.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Edit a parameter and shared list of values. Changes to the list of values could affect other reports.

Name: Country Type: String List of Values: Dynamic

Prompt Group Text: Please select Countries

Choose a Data Source:

☐ New ☒ Existing

Country - Prompt Group

Country [RQA1-BV]

Country - Prompt Group

Country [RQA1SQL-BV]

Country - Prompt Group

CountryName [RQA3-BV]

CountryName - Prompt Group

CREW_ID_VPC [FMMS-BV]

CREW ID VPC - Prompt Group

Value	Description
COUNTRY_NAME	COUNTRY_NAME

Important: If prompt group exists, choose that option (as on screenshot below) to avoid creation of duplicate values of the same prompt.

Edit a parameter and shared list of values. Changes to the list of values could affect other reports.

Name: Country Type: String List of Values: Dynamic

Prompt Group Text: Please select Countries

Choose a Data Source:

☐ New ☒ Existing

Country - Prompt Group

Country - Prompt Group

Country [RQA1-BV]

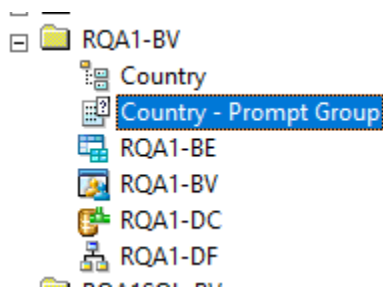
Country - Prompt Group

Country [RQA1SQL-BV]

Country - Prompt Group

Value	Description
COUNTRY_NAME	COUNTRY_NAME

After you added dynamic prompt to your report, you have completed manual creation of your Business Views in DEV environment, which you can export from DEV environment using RDM DevOps **Deploy to DEV** functionality.



For Business Views, RDM DevOps is actually **exporting** export.lcmbar file from manually created **DEV environment**, to be checked in source control and used to **import automatically** to other environments: **POC, TST, UAT, and PROD**.

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Creating Business Views in DEV is **always** manual process, created by developers.

Deploying Business Views in RDM DevOps

Note: Deploying label to DEV environment is actually exporting manually created business views into export.lcmbiar file, which needs to be copied to source control BV component folder.

Once export.lcmbiar file is exported from DEV and checked in source control, apply label for version, which is going to be used to import that export.lcmbiar file to other environments: POC, TST, UAT and PROD.

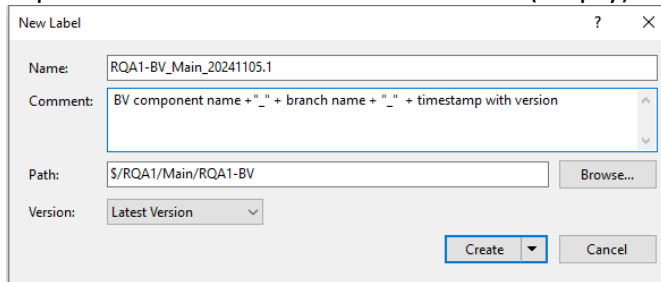
Task Summary

- 1- Label the business view folder in Azure DevOps.
- 2- Define a build name in RDM DevOps include only the business view.
- 3- Deploy Build to the DEV environment. export.lcmbiar file is created for the business view.
- 4- Copy export.lcmbiar file to the BV folder
- 5- Label the BV folder again.
- 6- Define a new build name in RDM DevOps to include the business view and any other components.
- 7- Deploy the build to any environment.

Detailed Steps

1- Label the business view folder in Azure

- a. Open Visual Studio and create label for (empty) business view folder



2- Define a build name in RDM DevOps to include only the business view.

- a. Select **Define Build Name** from the menu under the **Dev Lead's Task**. It will bring you to the **Define Release Version for System** page shown below.

Dev Lead's Task

Create Dev Environment
Create POC Environment
[SCM Structure](#)
[Define Build Name](#)
[Deploy to Dev](#)

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Select label that you previously created to export your business view from DEV. Click **Save**.

Define Release Version for System: RQA1

Build Name: RQA1 Version . . . ?

Import from Build Name: RQA1 Version 1.0.0.3

Reference Type: ChangeOrder

Change Order Number: 15465 ?

Component Builds:

Component Name	TFS label
RQA1-BV	RQA1-BV_Main_20241105.1

No records to display.

- 3- **Deploy Build to the DEV environment. An export.lcmbar file is created for the business view and an email with the link will be sent in the build result email by RDM DevOps.**

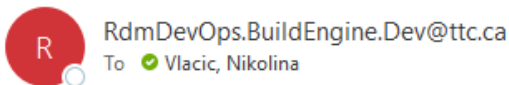
Deploy to DEV for System: RQA1

Release Version: RQA1 Version 1.0.0.4

☐ Show All

Component Name	TFS label
RQA1-BV	RQA1-BV_Main_20241105.1

RDMDevOps Notification - Build and Deployment Log - RQA1 (DEV)



Build and Deployment Log: RQA1 (DEV)

Actor: nvlacic

Build Server: D0195W003

Build Type: By Version

Build Name: RQA1 Version 1.0.0.4

Change Order Number: 15465

Details:

Build Label(s):

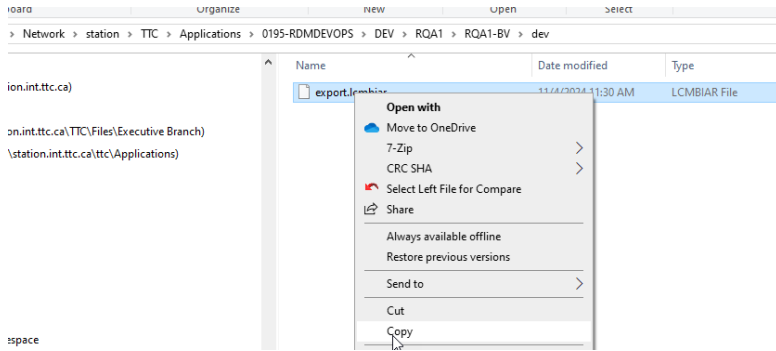
RQA1-BV_Main_20241105.1

- Export Business Views to [\\station\TTC\Applications\0195-RDMDEVOPS\DEV\RQA1\RQA1-BV\dev](#)
- Deployed RQA1-BV to D0073w001

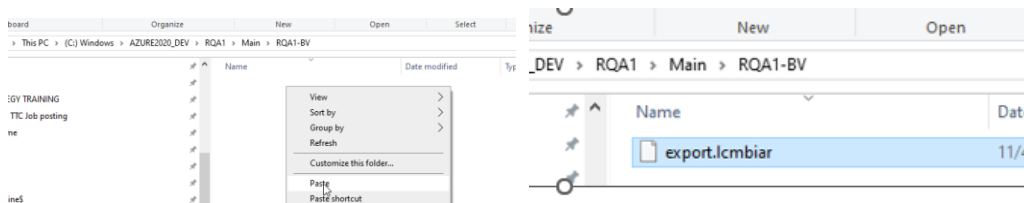
- 4- **Copy export.lcmbar file to the XXX-BV folder in your Azure BV component folder.**

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

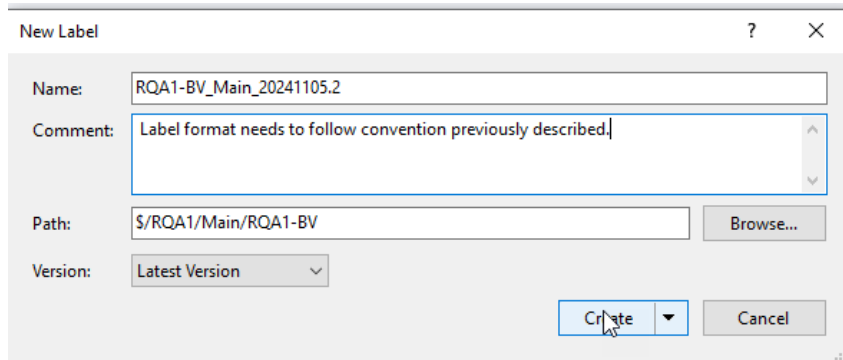
Open link from email and copy export.lcmbar file



Paste file into your workspace BV component folder.



Add export.lcmbar file (if not there) to your component folder, check in and label folder with new version:



USING BOE 4.3 WITH RDM DEVOPS AND AZURE

5- Define a new build name in RDM DevOps and include the business view.



RDMDevOps
Release and Deployment Manager for Azure DevOps

Welcome **INTTTC**

My System | Initiate System | Reports | System Search | Pending Tasks | User Guide.. | Admin Tools

Define Release Version for System: RQA1

Build Name: RQA1 Version ?

Import from Build Name: ▼

Reference Type: ▼

Change Order Number: ?

Component Builds:

Component Name	TFS label		
RQA1-BV	RQA1-BV_Main_20241105.2		
RQA1-CR	RQA1-CR_Main_20241105.1		

6- Deploy the build to the DEV or POC environment.

- Click on either Deploy to Dev or Deploy to POC link depending on where you would like to deploy your build.

Dev Lead's Task

Create Dev Environment
Create POC Environment
[SCM Structure](#)
[Define Build Name](#)
[Deploy to Dev](#)
[Deploy to POC](#)
[Ready for QA Testing](#)
[Ready for Production](#)

- Select your release version which is your build number and click **Save** to deploy.

Deploy to POC for System: RQA1

By Release Version | By Build Label | By Saved Build Setting

Release Version: ▼

☐ Show All

Component Name	TFS label
RQA1-BV	RQA1-BV_Main_20241105.2
RQA1-CR	RQA1-CR_Main_20241105.1

USING BOE 4.3 WITH RDM DEVOPS AND AZURE

Creating and Deploying a Universe in RDM DevOps

Create folder under Main branch named **<name>-UNI**. Define Universe Component by creating new component and selecting BOE Universe Component type. Create environments and add users for each environment by clicking on links to Manage users for each environment.

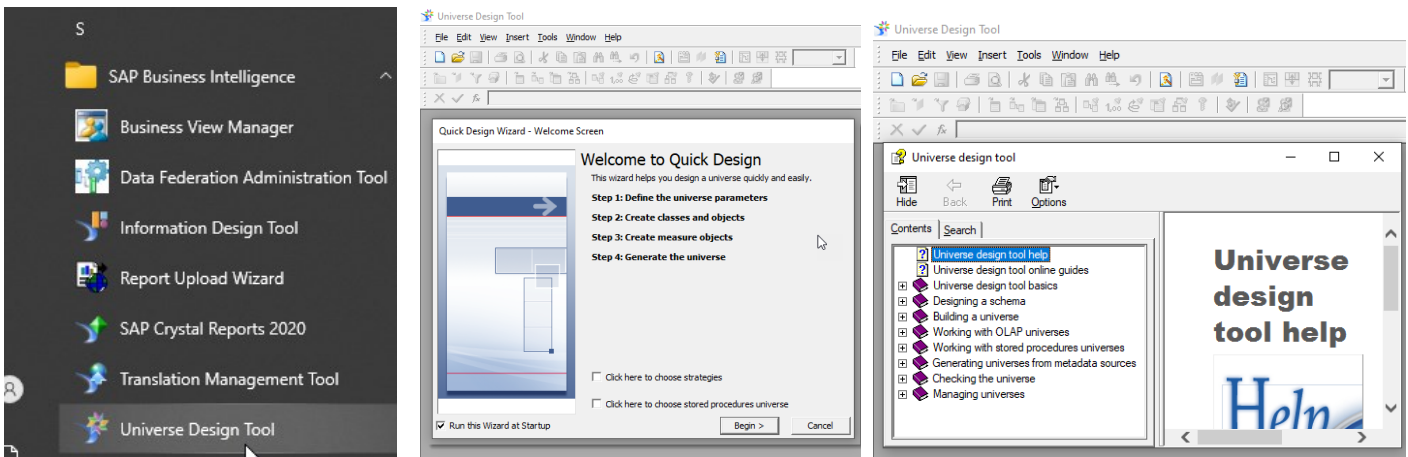
RDEV2 / RDMDEV2-UNI

[Manage Developers](#) [Manage POC Users](#) [Manage Testers](#) [Manage UAT Users](#) [Manage PROD Users](#)

Component Type: BOE Universe
Component Name: RDMDEV2-UNI
Release Analyst: BOEAdmin
Description: TEST UNI ON NEW BOE 4.3
Root Folder: RDMDEV2-UNI

Save Return

Deploying a universe in RDM DevOps is the same as deploying Business Views. Universe is manually created using SAP Universe Design Tool, included in SAP Business Intelligence Client. When logged in Universe Design Tool, click on F1 to get Universe Design Tool help.



Deploying universe component to DEV environment is actually exporting export.lcmbar file to be checked in to Azure component folder, labeled and deployed to other environments: POC, TST, UAT and PROD.

Deploy to DEV for System: RDEV2

By Release Version **By Build Label** By Saved Build Setting

Import from Build Name: RDEV2 Version 1.0.0.2
[Import](#)

Reference Type: ChangeOrder
Change Order Number:

☐ Save Setting

[+ Add new record](#)

Component Name	TFS label
RDMDEV2-UNI	RDMDEV2-UNI_Main_20241030.1

Save Cancel



USING BOE 4.3 WITH RDM DEVOPS AND AZURE

You will use this label and define a build with it. Please refer to the section Deploying of Business Views as a guide.

For example, to deploy to POC, click Deploy to POC link. Email is received to notify if deployment was successful.

Dev Lead's Task

Create Dev Environment
Create POC Environment
[SCM Structure](#)
[Define Build Name](#)
[Deploy to Dev](#)
[Deploy to POC](#)
[Ready for QA Testing](#)
[Ready for Production](#)

RDMDevOps Notification - Build and Deployment Log - RDEV2 (POC)



RdmDevOps.BuildEngine.Dev@ttc.ca
To: Vlacic, Nikola

Build and Deployment Log: RDEV2 (POC)

Actor: nvlacic
Build Server: D0195W003
Build Type: By Version
Build Name: RDEV2 Version 1.0.0.1
Change Order Number: 15287
Details:
Build Label(s):
RDMDEV2-UNI_Main_20241030.1
• Universe created on D0073w002. Access granted to RDMDEV2-UNIPOC.
• Deployed RDMDEV2-UNI to D0073w002

Common Email Error Messages for Deployment

Error Message

ODBC Connection not defined in RDM DevOps ...

Cause

ODBC Data Source name in report is not the same as the ODBC name in RDM .

Solution

Update the report to use the ODBC DSN in RDM DevOps or change the ODBC DSN name in the component.

Unable to find the database credentials in ARIS ...

All reports require the Environment, database, ODBC name and password saved in our ARIS database.

Contact a DBA and request the required information to be put into ARIS for all environments and same user ID as saved in report properties.

File Repository Server error ...

A BOE file input sever is down

Contact BOE Admin or Call service desk with the error message.

The following RPT files either do not contain a Document Title ...

Document Title was not entered in the summary of the BOE report.

Enter the proper document title format or enter a document title if none has been entered for the crystal report.

Unable to connect to cluster @BOE.INT.TTC.CA:6400 to retrieve updated CMS member list. Logon cannot continue.

When a server is rebooted the first deployment on the server after reboot gives this error.

Retry the build a second time.

Glossary**A**

Azure DevOps Microsoft's source code control software.

B

Business Intelligence Another term for ability to report/process data and provide business information in an understandable format.

Business Objects Enterprise(BOE) The engine that stores, organizes, and runs a variety of reporting types of files providing numerous tools.

Business Views In TTC it is mainly used for dynamic prompts.

C

Crystal Reports Reporting (business intelligence) software used to report on a database using a thick client.

D

Data Source Name The name of the ODBC connection that is created on a workstation/server.

DEV Development Environment

G

RDM DevOps TTC's in-house deployment web application.

L

Label A label in Azure DevOps stamps/snapshots/records a set of files and the version they are at when the label was created.

P

POC Proof of Concept Environment

PROD Production Environment

T

TNS Name The database alias name. Ex (AMOS_dev, AMOS_prod)

U

UAT User Acceptance Testing Environment

Universe A meta-data layer in between the database and the reporting software. Can be used in Crystal Reports and WEBI.

W

Web Intelligence (WEBI) A thin client reporting (business intelligence) software used to report on a database which is configured in a universe.